1.2. Transferring images to the PC

Now that we know something about image formats, let's look at the devices which are used to capture/store these images.

1.2.1. Digital Cameras

A digital camera is a camera that captures still photographs digitally by recording images on a light-sensitive sensor.

Digital cameras include features that are not found in film cameras, such as displaying an image on the camera's screen immediately after it is recorded, the capacity to take thousands of images on a single small memory device, the ability to record video with sound, the ability to edit images, and deletion of images allowing re-use of the storage device. Digital cameras are incorporated into many devices ranging from PDAs and mobile phones (called camera phones) to vehicles.

The term digital still camera (DSC) usually refers to a live-preview digital camera, with an electronic display, usually a rear-mounted LCD as the principal means of framing and previewing, before taking the photograph and for viewing stored photographs. All use either charge-coupled device (CCD) or a CMOS image sensor to sense the light intensities across the focal plane.

Images may be transferred to a computer, printer or other device in a number of ways—the USB mass storage device class makes the camera appear to the computer as if it were a disk drive, the Picture Transfer Protocol (PTP) and its derivatives may be used, FireWire is sometimes supported, and the storage device may simply be removed from the camera and inserted into another device.

The resolution of a digital camera is often limited by the camera sensor (typically a CCD or CMOS sensor chip) that turns light into discrete signals, replacing the film in traditional photography. The sensor is made up of millions of "buckets" that essentially count the number of photons that strike the sensor. This means that the brighter the image at that point, the larger the value that